

Assignment 4

RDF/XML and Schema “Debugging”

You are given an RDF Schema and an RDF/XML data fragment describing a small HealthSense234 dataset.

RDF Schema (in Turtle)

- `hs:Patient rdfs:subClassOf hs:Person`
- `hs:Doctor rdfs:subClassOf hs:Clinician`
- `hs:Clinician rdfs:subClassOf hs:Person`
- `hs:hasVisit`
 - `rdfs:domain hs:Patient`
 - `rdfs:range hs:Visit`
- `hs:hasAttendingDoctor`
 - `rdfs:domain hs:Visit`
 - `rdfs:range hs:Doctor`
- `hs:livesIn`
 - `rdfs:domain hs:Person`
 - `rdfs:range hs:City`

RDF/XML Fragment

```
<rdf:Description rdf:about="hs:Visit1">
  <hs:livesIn rdf:resource="hs:Cleveland" />
  <hs:hasAttendingDoctor rdf:resource="hs:City1" />
</rdf:Description>
<rdf:Description rdf:about="hs:Alice">
  <rdf:type rdf:resource="hs:City" />
  <hs:hasVisit rdf:resource="hs:Visit1" />
</rdf:Description>
<rdf:Description rdf:about="hs:Bob">
  <rdf:type rdf:resource="hs:Patient" />
  <hs:hasAttendingDoctor rdf:resource="hs:DrSmith" />
</rdf:Description>
<rdf:Description rdf:about="hs:DrSmith">
  <rdf:type rdf:resource="hs:Doctor" />
</rdf:Description>
```

Identify Violating Triples

List all triples that violate the intended schema constraints.

✓ Answer ✓

The first `livesIn` is a domain constraint, it can't exist on a visit.

The first `hasAttendingDoctor` is likely also wrong, unless the doctor is named `City1`

Alice is most likely not of type `City`, but if she is a city, she can't `hasVisit` ed something as a city.

The second `hasAttendingDoctor` can only exist on visits, not patients.

Explain Violations

For each violating triple, identify which schema rule (domain, range, subclass) is violated. Briefly explain why.

✓ Answer

Domain, range, subclass, domain.

`livesIn` can't exist on Visits.

Cities aren't valid doctors for `hasAttendingDoctor`.

Alice is either a city or a patient, but not both.

`hasAttendingDoctor` can't exist on patients.

Propose corrections

For each violating triple, propose a corrected version of the triple.

```
<rdf:Description rdf:about="hs:Visit1">
  <hs:hasAttendingDoctor rdf:resource="hs:DrSmith" />
</rdf:Description>
<rdf:Description rdf:about="hs:Alice">
  <rdf:type rdf:resource="hs:Patient" />
  <hs:hasVisit rdf:resource="hs:Visit1" />
</rdf:Description>
<rdf:Description rdf:about="hs:Bob">
  <rdf:type rdf:resource="hs:Patient" />
  <hs:livesIn rdf:resource="hs:Cleveland" />
</rdf:Description>
<rdf:Description rdf:about="hs:DrSmith">
  <rdf:type rdf:resource="hs:Doctor" />
</rdf:Description>
```

Knowledge Graph and Ontology Design

You are given a small knowledge graph describing people, roles, and locations in the HealthSense234 Lab scenario.

Knowledge Graph (KG): Each fact is given as a triple $r(x, y)$.

```
siblingOf(Alice, Ben)
parentOf(Ben, Chloe)
type(Chloe, Nurse)
subClassOf(Nurse, Employee)
subClassOf(Employee, LabAuthorizedPerson)
livesIn(Alice, Columbus)
livesIn(Chloe, Cleveland)
livesIn(David, Pittsburgh)
locatedIn(Columbus, Ohio)
locatedIn(Cleveland, Ohio)
locatedIn(Pittsburgh, Pennsylvania)
storedIn(PrototypeX, MedLab)
seenAt(Alice, MedLab)
```



A partial Ontology O is given, and currently contains the following classes and relationships.

Classes

- Person
- Location
- City
- State
- Facility

Subclass relationships

- City isSubClassOf Location
- State isSubClassOf Location

Property domain/range constraints

- livesIn
 - domain: Person
 - range: Location
- locatedIn
 - domain: Location
 - range: Location
- seenAt
 - domain: Person
 - range: Facility

- storedIn
 - domain: Artifact
 - range: Facility

a

From the KG, list all the nodes that correspond to classes already defined in O.

✓ Answer

Person:

- Alice
- Bob
- Ben
- Chloe
- David

Location/City/State:

- Columbus
- Pittsburgh
- Cleveland
- Ohio
- Pennsylvania

Facility:

- MedLab

b

Extend O so that it properly “covers” the KG. Your extension should include missing classes, appropriate subClassOf relationships, and any necessary typing for entities. You may present your answer as a list of new classes and subclass relationships, or a small diagram, or use RDF/XML or Turtle code.

✓ Answer

All associations given are added to this turtle code. An assumption we made is that LabAuthorizedPerson is a Person.

```
%% Person, Location, City, State, Facility %%
```

```
- hs:City rdfs:subClassOf hs:Location
- hs:State rdfs:subClassOf hs:Location
- hs:Nurse rdfs:subClassOf hs:Employee
```

- `hs:Employee rdfs:subClassOf hs:LabAuthorizedPerson`
- `hs:LabAuthorizedPerson rdfs:subClassOf hs:Person`
- `hs:livesIn`
 - `rdfs:domain hs:Person`
 - `rdfs:range hs:Location`
- `hs:locatedIn`
 - `rdfs:domain hs:Location`
 - `rdfs:range hs:Location`
- `hs:seenAt`
 - `rdfs:domain hs:Person`
 - `rdfs:range hs:Facility`
- `hs:storedIn`
 - `rdfs:domain hs:Artifact`
 - `rdfs:range hs:Facility`
- `hs:siblingOf`
 - `rdfs:domain hs:Person`
 - `rdfs:range hs:Person`
- `hs:parentOf`
 - `rdfs:domain hs:Person`
 - `rdfs:range hs:Person`

C

Assign Types (Classes) to the following nodes with proper RDFS declaration:

Alice, Chloe, Columbus, MedLab, PrototypeX.

```
hs:Alice rdf:type hs:Person
hs:Chloe rdf:type hs:Nurse
hs:Columbus rdf:type hs:City
hs:MedLab rdf:type hs:Facility
hs:PrototypeX rdf:type hs:Artifact
```

Knowledge Graph Reasoning

Using the KG and ontology from Q2, answer the following questions.

Assume we have the following three inference rules, given the ontology:

- Subclass inheritance: If `subClassOf(A, B)` and `type(x, A)`, then `type(x, B)`.
- Transitivity (`locatedIn`): If `locatedIn(x, y)` and `locatedIn(y, z)`, then `locatedIn(x, z)`.
- Symmetry (`siblingOf`): If `siblingOf(x, y)`, then `siblingOf(y, x)`.

Fact checking

for each statement/triple below, can it be inferred by applying only the three rules to the KG? Mark

“Yes” or “No”

The “No” Nichols

Statement	Answer
type(Chloe, Employee)	Yes
type(Chloe, LabAuthorizedPerson)	Yes
locatedIn(Alice, Ohio)	No
locatedIn(David, Ohio)	No
siblingOf(Ben, Alice)	Yes
type(Alice, Employee)	No
locatedIn(Chloe, Pennsylvania)	No

Explain Inference

Choose two statements marked as inferable, and explain: which facts (triples) are used, and which rules are applied, to derive the statements?

✓ Answer

```
type(Chloe, Nurse)
subClassOf(Nurse, Employee)
→ type(Chloe, Employee) (Subclass Inheritance)

siblingOf(Alice, Ben)
→ siblingOf(Ben, Alice) (Symmetry)
```

Detective Reasoning

A prototype was stolen from HealthSense234 MedLab. The current crime investigation collects the following clues describing the suspect:

- Lives in Ohio
- Is related to Ben
- Has Lab Authorization (an “insider”)
- Was seen at MedLab

Who could be the most likely suspect? Justify your answer using the KG and inference rules. Note that there may be “missing” fact in the KG that may or may not be inferred by using the inference rules alone. For example, no property called seenAt between X and Y does not necessarily mean X was not at the location Y.

Further investigation suggests a new fact: type(Alice, LabAuthorizedPerson) – “we found Alice actually has gained authorized access to Lab”. How would your answer change? Briefly explain.

✓ Answer

Lives in Ohio: Alice, Chloe
Related to Ben: Alice, Chloe
Is authorized: Chloe
Seen at MedLab: Alice

I would say Chloe is the most likely as she satisfies 3/4 criteria, and would be harder to steal it if she were not authorized, but seen at medlab instead.

With this new information, I would say Alice, as she would satisfy 4/4 criteria, making her the most likely to have stole it.